Risk and Benefits of Digitized Money and Assets

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This course will cover theoretical models examining potential risk channels generated by CBDC and cryptomarket developments.

The required background includes training from first-year graduate macroeconomics: e.g., Lagrangian calculus-based models for both competitive and monopolistic models, with an understanding of general equilibrium, firm optimization, consumer optimization, and the ability to read first-order conditions.

Knowledge of macroeconomic simulation and calibration methods is helpful, but not required. No knowledge of cryptocurrency markets, DeFI, CBDCs, banking, or monetary policy is assumed.

Topics and Core Papers:

(1) CBDC and Bank Stability
   a. Andolfatto, David. Assessing the impact of central bank digital currency on private banks. *Economic Journal*, 131(634):525–540, 2021. URL: https://academic.oup.com/ej/article/131/634/525/5900973
   b. Chiu, Jonathan, Seyed Mohammadreza Davoodalhosseini, Janet Jiang, and Yu Zhu. 2023. "Bank Market Power and Central Bank Digital Currency: Theory and Quantitative Assessment." *Journal of Political Economy* 131 (5): 1213–48. https://doi.org/10.1086/722517.
   c. Whited, Toni M. and Wu, Yufeng and Xiao, Kairong, Will Central Bank Digital Currency Disintermediate Banks? (June 13, 2023). Available at SSRN: https://ssrn.com/abstract=4112644 or http://dx.doi.org/10.2139/ssrn.4112644
   d. Bian, Wenlong, Yang Ji, and Peng Wang. 2021. "The Crowding-out Effect of Central Bank Digital Currencies: A Simple and Generalizable Payment Portfolio Model." *Finance Research Letters* 43 (November): 102010. https://doi.org/10.1016/j.frl.2021.102010.
   e. Li, Jiaqi. 2023. "Predicting the Demand for Central Bank Digital Currency: A Structural Analysis with Survey Data." *Journal of Monetary Economics* 134 (March): 73–85. https://doi.org/10.1016/j.jmoneco.2022.11.007.
   f. Li, Jiaqi, Andrew Usher, and Yu Zhu. 2023. "Central Bank Digital Currency and Banking Choices." Working paper, September 22. https://doi.org/10.2139/ssrn.4705512.

g. Nyffenegger, Remo. 2024. “Central Bank Digital Currency and Bank Intermediation: Medium of Exchange vs. Savings Vehicle.” *European Economic Review* 170 (November): 104890. https://doi.org/10.1016/j.euroecorev.2024.104890.
h. Williamson, Stephen D. 2022. “Central Bank Digital Currency and Flight to Safety.” *Journal of Economic Dynamics and Control* 142 (September): 104146. https://doi.org/10.1016/j.jedc.2021.104146.
i. Brunnermeier, Markus K., and Dirk Niepelt. 2019. “On the Equivalence of Private and Public Money.” *Journal of Monetary Economics*, SPECIAL CONFERENCE ISSUE: “Money Creation and Currency Competition” October 19-20, 2018 Sponsored by the Study Center Gerzensee and Swiss National Bank, vol. 106 (October): 27–41. https://doi.org/10.1016/j.jmoneco.2019.07.004.

(2) CBDC Macro-Risk Consequences

a. Barrdear, John, and Michael Kumhof. 2022. “The Macroeconomics of Central Bank Digital Currencies.” *Journal of Economic Dynamics and Control*, The Economics of Digital Currencies, vol. 142 (September): 104148. https://doi.org/10.1016/j.jedc.2021.104148.
b. Davoodalhosseini, Seyed Mohammadreza. 2022. “Central Bank Digital Currency and Monetary Policy.” *Journal of Economic Dynamics and Control* 142 (September): 104150. https://doi.org/10.1016/j.jedc.2021.104150.
c. Williamson, Stephen. 2022. “Central Bank Digital Currency: Welfare and Policy Implications.” *Journal of Political Economy* 130 (11): 2829–61. https://doi.org/10.1086/720457.
d. Fernández-Villaverde, Jesús, Daniel Sanches, Linda Schilling, and Harald Uhlig. 2021. “Central Bank Digital Currency: Central Banking for All?” *Review of Economic Dynamics*, Special Issue in Memory of Alejandro Justiniano, vol. 41 (July): 225–42. https://doi.org/10.1016/j.red.2020.12.004.
e. Schilling, Linda, Jesús Fernández-Villaverde, and Harald Uhlig. 2024. “Central Bank Digital Currency: When Price and Bank Stability Collide.” *Journal of Monetary Economics* 145 (August): 103554. https://doi.org/10.1016/j.jmoneco.2024.01.007.
f. Jiang, Janet, and Yu Zhu. 2021. “Monetary Policy Pass-Through with Central Bank Digital Currency.” *Bank of Canada Staff Working Paper 2021-10*.

(3) Stablecoins: Financial Risk

a. Brunnermeier, Markus K., and Dirk Niepelt. 2019. “On the Equivalence of Private and Public Money.” *Journal of Monetary Economics*, SPECIAL CONFERENCE ISSUE: “Money Creation and Currency Competition” October

19-20, 2018 Sponsored by the Study Center Gerzensee and Swiss National Bank, vol. 106 (October): 27–41. https://doi.org/10.1016/j.jmoneco.2019.07.004.

b. Williamson, Stephen. 2024. "Deposit Insurance, Bank Regulation, and Narrow Banking." *Journal of Economic Theory* 219 (July): 105859. https://doi.org/10.1016/j.jet.2024.105859.

c. Lyons, Richard K., and Ganesh Viswanath-Natraj. 2023. "What Keeps Stablecoins Stable?" *Journal of International Money and Finance* 131 (March): 102777. https://doi.org/10.1016/j.jimonfin.2022.102777.

d. Hui, Cho-Hoi, Andrew Wong, and Chi-Fai Lo. 2025. "Stablecoin Price Dynamics under a Peg-Stabilising Mechanism." *Journal of International Money and Finance* 152 (March): 103280. https://doi.org/10.1016/j.jimonfin.2025.103280.

e. Li, Ye, and Simon Mayer. 2025. Financial Intermediation with Fragile Seigniorage: A Dynamic Model of Stablecoin Issuers. https://www.dropbox.com/scl/fi/q7hjps4i5wt0rioxgfsmt/Stablecoin.pdf?rlkey=3upcyo9zfxti3fxggj7mkpb0f&e=1&st=9iwkkvqu&dl=0

f. Gorton, Gary B., Elizabeth C. Klee, Chase P. Ross, Sharon Y. Ross, and Alexandros P. Vardoulakis. 2025. "Leverage and Stablecoin Pegs." *Journal of Financial and Quantitative Analysis*, March 14, 1–38. https://doi.org/10.1017/S0022109025000134.

(4) Open-Economy Currency Competition

a. Benigno, Pierpaolo, Linda M. Schilling, and Harald Uhlig. 2022. "Cryptocurrencies, Currency Competition, and the Impossible Trinity." *Journal of International Economics*, NBER International Seminar on Macroeconomics 2021, vol. 136 (May): 103601. https://doi.org/10.1016/j.jinteco.2022.103601.

b. Ferrari Minesso, Massimo, Arnaud Mehl, and Livio Stracca. 2022. "Central Bank Digital Currency in an Open Economy." *Journal of Monetary Economics* 127 (April): 54–68. https://doi.org/10.1016/j.jmoneco.2022.02.001.

c. Azzimonti, Marina, and Vincenzo Quadrini. 2024. "Digital Assets and the Exorbitant Dollar Privilege." *AEA Papers and Proceedings* 114 (May): 153–56. https://doi.org/10.1257/pandp.20241069.

(5) Crypto Monetary Theory: Three Frameworks, Positive and Normative

a. Jermann, Urban J. 2024. "A Macro Finance Model for Proof-of-Stake Ethereum." Working paper, February 7. https://doi.org/10.2139/ssrn.4335835.

b. Li, Ye, and Simon Mayer. 2022. “Money Creation in Decentralized Finance: A Dynamic Model of Stablecoin and Crypto Shadow Banking.” SSRN Scholarly Paper No. 3757083. Social Science Research Network, April 8. https://doi.org/10.2139/ssrn.3757083.
c. Jermann, Urban, and Haotian Xiang. 2025. “Tokenomics: Optimal Monetary and Fee Policies.” *Journal of Monetary Economics* 155 (October): 103808. https://doi.org/10.1016/j.jmoneco.2025.103808.